

ActInf GuestStream 077.1 ~ Gareth Lee Stubbs: "The Police Hunch" Bayes

GuestStream | Gareth Lee Stubbs: Bayesian Brain, Active Inference | 1:21:28

hello and welcome everyone this is
active inference guest stream
77.1 we're here with Gareth stubs going
to be discussing the police hunch so
Gareth thank you very much for joining
looking forward to your presentation and
the discussion we'll
have thank you so much for having me
Daniel and um it's an absolute pleasure
to be on the stream one that I've
watched many times um and it's
privileged to be able to talk through
the paper um I'm hoping to give you a
little bit of background as to why I
wrote the paper where I've come from a
little bit of History talk through the
paper itself and then hopefully take
some interesting questions um if there's
anybody watching or um from Daniel
himself who's always got some uh very
insightful questions to ask um uh so if
you're all right Daniel am I okay to
start and get going yeah okay so um a
little bit of background so I'm a
relatively new academic um I was a
police officer for 18 years uh in the UK
I was in a Northern English Force um and
during my career uh I did a wide range
of stuff from like Frontline emergency
response blue lights to uh community
policing which is building relationships
with the Community Gathering
intelligence and then um uh working on
long-term problem solving through to
working in uh National strategy uh I

worked for the College of policing and
um more laterally the home office at the
inspectorate and the inspectorate was an
organization where we would inspect
police officers and make sure that um uh
the standards of policing and how
they're recorded and monitored and all
that was up to a particular level and
for the last uh five years um of my
policing career I was studying a
part-time PhD uh on the side and you
would think uh with me having written
this paper it would be in a police
decision making or similar and it and it
wasn't I actually studied um uh using
two theories which were uh
representative bureaucracy and social
embeddedness uh and it was actually
about how uh new police recruits uh
navigated the Police recruitment process
via the use of their social network
um it sounds complicated and convoluted
and it was um I essentially spent a lot
of time uh trying to figure out how
people gained advantage or disadvantage
through their current social networks
whilst trying to get into the police uh
with the Hope um of uh figuring out uh
why our um uh recruitment process was so
discriminatory um and and I've I've been
working on publishing from that PhD uh
I've only just finished actually uh over
this last maybe two three years uh
whereupon I went out to the Middle East
to teach police officers now and there's
two people that I want to discuss during
this stream the first being um uh two
different uh groups of people the first

one being Professor friston who was just amazing when I contacted him uh and the other being two uh very well-known criminologists called uh Ken peas um and Jason roach and this entire like uh lane or path of study came out of a a discussion with them because I was talking about um very strange decision making that you make while in service so you make decisions that other people look at and say oh that's a very strange decision why did that police officer make that decision but when you speak to the officer they're often able to relate quite a complicated series of past experiences that have led to the decision that um they've made uh and it was K who's again a very well-known criminologist who said to me look you need to read about the Basi and brain he said look this what you're describing is is how the Bas and brain works you know um so he set me on this kind of Down The Rabbit Hole uh Journey if you like um and did a lot of uh reading of the foundational papers around the basing brain which I found very complicated because I didn't have a background in uh mathematics or Neuroscience or cognitive science so it was kind of like jumping in the deep end um but he recommended a really good book called Super forecasters by tlock um some of you may have heard of it some of you may not have heard about it but um it's essentially the human art of prediction like how intuition is made in

the basian brain and um I found the book revelatory because uh when I look back at my police experience I could see the development of my basing brain as time went on and how my decision- making changed and this was like kind of mixed because we all have our backgrounds and our history but when I was at the home office in London and we were looking at the way that police forces were organized and governed and all that sort of thing we could see that there was a general tendency towards this kind of promotion of the generalist so a lot of your senior managers seem to have been able to like Bounce Around the organization they've done lots of different things some have been detectives some have been uniformed some have been um uh in corporate some have been Firearms you had all these different like kind of um disciplines if you like um and you ended up with a lot of senior leaders who were jack of all trades and essentially uh masters of non and this was in direct contrast to a lot of the research that looked at you know teams of Specialists people with specialist knowledge that have built up very very solid what you could call like Basi and building blocks when it comes to specialist decision- making and then when you put them together the matching of all these specialisms then resulted in higher performance of external private companies so we were we were trying to think about how this might look in policing and it was only when I

wrote H Read Super forecasting solid blocks of prediction that I started to understand how the brain may work now of course uh if you at any point uh going to YouTube and you type in the basing brain you're going to get professor frisen in some way um so that was my first introduction and this was about two years ago and I absolutely devoured um all of his podcasts I mean everything that I could find find um and that kind of linked in a lot with the llm development obviously we had the side by side active inference seemed to be like picking up steam like crazy and then we had the llms and the AI research picking up steam and then you could see how the two were starting to like twist and merge and there was some really good podcasts from uh machine learning Street talk and and um Kurt J mongal uh podcast Theory of Everything um well we're talking really long form like two and a half three hour discussions on this stuff and I was just absolutely immersed and I started to think about how this has never been used in practice like in policing you know like a lot of the time we go to these training sessions you'd have some experienced practitioners talking to you about how they made decisions we'd have decision-making models and me with my relatively new academic brain was immediately going straight to Google Scholar and saying okay what's the empirical basis for these decision-making models I know everyone's using them but do we know if

they work do we know if they speed up
our decision making are they better with
those who've got better basing building
blocks are they not better with so and
when you start to look at these
questions you realize that there's
almost no research um uh to support
police decision- making models um uh the
best stuff that I found on that is in
Andy Clark's book um The Experience
machine there a full chapter on looking
at police decision making um uh anyway
I'm getting slightly distracted how did
I actually come to the paper so um there
was one experience in my service and I'm
going to tell the story because it's in
the paper if that's all right Daniel is
that

okay okay um so I was a newly promoted
sergeant and that's like first line
supervisor in the police I had a team of
about 15 police officers and I was
probably about 27 28 years old um and I
was out with one of those police
officers who was new and he was trying
to build his confidence so I said to him
look um Jamie I said we're going to go
to all the bad stuff tonight so when
when the nasty incidents come in we're
going to go and I'm going to see if I
can work with you on building your
confidence in these sort of like you
know these these high-risk situations
and he was all up for it
uh and then a domestic violence incident
came in and the male um offender was
still inside the um house and we knew
that there had been violent shoes

because the person who rang it in was a neighbor and the neighbor had said that she was severely injured and bleeding so uh because there was a a chance that it could get rough and we might need to um make an arrest straight away I thought this would be good for Jamie to go to uh and he was a taser officer and I don't know how how like uh experien you guys are so I'll just let you know but us police officers in England we don't carry guns like um there's probably about five or six% of police officers that carry guns at the most and a lot of them are only sent to incidents that may involve Firearms so the vast majority of incidents we go to we have a Baton which is like a stick um and a uh a spray uh and there're the they're the two bits that we've got to deal with it now Jamie had just got his taser um and uh he was trying to build his confidence um so we turned up there was a very injured lady on the sidewalk she was pulling clumps out of her hair she'd been clearly punched a great deal her face was an absolute mess and she told us that he was still inside the male offender uh and that obviously he will be violent towards us as police officers now that yeah if you read the paper this may come across as if it was a like a kind of um a one-off incident it wasn't you go to things like this all the time on a busy Friday or Saturday night I could go to the two or three of these maybe um so this was very very normal nothing out of the

ordinary and I said to Jamie okay well
let's go on um we'll wait for some back
up because he might he might be violent
this guy and you know we'll go in and
we'll we'll see what we can do he's
obviously going to need to be arrested
because of the damage that he' done to
the
lady um and uh she said no you I've got
my one-year-old upstairs and he's in bed
and the minute that she said that like
we can't wait for backup we have to go
in because there's a chance that uh the
the child could be
injured so obviously that changed our
decision- making and as we're working up
the front door up to the front door the
door opens and this big muscly guy comes
to the front door
and the thing that got me at that time
and again I just have to stress like I
got these were very very normal
incidents like would go to these all the
time there was something about this
incident where I felt like my heart hit
my stomach big adrenaline dump all of
the um uh prickles on my arm stood up
like my hair stood up power erection and
up the back of my neck and I remember
looking at Jamie Jamie didn't feel this
and I spoke to him afterwards it was
really strange because there was two of
was presented with the same incoming
sensory stimuli I had a very different
bodily reaction um and we talked about
it afterwards I mean at the time out
like I say I was eight or nine years of
experience Jamie was like coming up to

two so there was a big buildup of basing
priors in in me for some reason um
anyway uh let's qu long story short uh
we ended up fighting with the male uh he
was very violent towards us really
really bad incident um and when we
finally got him in handcuffs as we got
to the front door we noticed that he
dropped a knife and he'd had a knife
when I'd pulled him out of the
doorway now um this is like one of those
moments that you have in the policing
career where you're like your mortality
punches you in the face you know because
had we gone inside or had we engaged him
in a very different way there's a chance
that both of us would not have come away
from that incident and the sometimes
that like hits you and it hit us both
then and and Jamie got very upset
remember it very distinctly we we walked
into the kitchen we had to go and check
that the young lad was okay and he was
and obviously we got the injured party
back in to the
house uh but I remember doing like an
immediate reflection afterwards and
saying to Jamie like I have never had
this severity of body bodily
reaction like
that that came on so quickly and I've
used the word salans to describe it in
the paper but there was something that
was distinctly different and I don't
know what it was but obviously whatever
signals I was receiving be it from the
lady on the curb or from the M's
Behavior as we approached the door there

was something in me that told me
interceptive so it was an internal
feeling that then triggered my ctive
processes to approach the incident
differently anyway that incident
obviously he went to court he was he
went to jail actually and so the
incident itself we managed to safeguard
the lady in the sun which is always a
good outcome um but for me it really
stuck with me uh for a couple of reasons
the first was the the bodily reaction
that was so different was something that
Jamie didn't receive at all so there was
no difference in him he he saw nothing
um of the evidence that I saw and when I
say saw this is obviously um and again
cover it in the paper there's so much
information that you're taking in these
incidents and it's in such a temporal
window that such a tight temporal window
that your ability to be able to affect
the environment that you're in is uh
very very highly constrained because
you're only going to be at that incident
maybe for minutes if you're lucky an
hour or over depending on how busy the
radio is so you can't do what you would
call any longitudinal Niche Niche
construction your ability to mold or
shape your environment when you turn up
as a police officer is limited solely to
the time that you're there and because
you don't know any of the people
there there's something going on in the
police officers brain and again this is
discussed in the paper that is very
different to a normal work environment

so this is I'm just going to I I'll try
my best to give you an example so if you
work on a till in customer services you
have a very predictable way of living
your your day out and of course the
unpredictability comes from the
customers but there's often a set of
like procedures or protocols that you
follow that enable you to deal with
those people in a particular way and
allows you to predict your work and
hopefully comfortably do that now when
you're a police officer you have the law
and this was interesting actually Daniel
because when we first had a chat about
this I think you put me in touch with
somebody who was talking about the law
as an agent of active inference which is
proper interesting stuff and if we get
time like later on it might be worth a
discussion but the officers are working
within the law so you've got uh bounds
if you like you have boundaries that you
work within and
um there's particular actions you can
take but everything is so highly
unpredictable I'm not working with a
till I'm working directly with people
it's often in a very very emotionally
charged environment because that's why
police get called in the first place you
don't know any of the people that you're
dealing with and they're not coming to
you asking for the same service so it's
not like you're going up to a shop and
saying I want to buy this again and
again and again which is the same thing
that you'd get as a cashier when we turn

up a lot of the time people don't even want police there so you come in as like a kind of alien if you like and there's often something that again it's like high emotions everything's going a little bit crazy decision making of those people is normally totally out of the norm you're trying to bring everything back within the norm so if you like that the officer is like an agent of allostasis we're trying to bring back homeostasis to the environment when you get there but it's so messy and so difficult that you often end up making uh decisions other people may look at and say why have they done that that's such a strange decision but if you've been and and I suppose in the paper and I go through it if you've been in that particular situation a number of times before and you have a good set of building blocks that allow you to make improved decision-making as a result of that experience you may end up with improved outcomes and I'm not going to say that you will but you may and it's a solid Theory to um uh back up good experience Le decision making and there's lots of things that you can look at and hopefully the paper towards the end might be able to just talk about some of those and I can take the viewers through them so that's the kind of background the salience of this particular incident um is the one that the paper is based on it was so different to me during my police career that I needed something that explained

it this Theory explains it in a lovely way it's so nice um and obviously the the kind of absorption of uh Carl Friston's um uh work over time and the reading of the books I mean that's inference book by Thomas P I know that you're doing a series on it a reading group on it it's brilliant I would say it's a little bit hard going don't pick it up as your first book do some listening to the podcast first try and get your head around how it all works and then pick up the book and go through it it gets a little bit heavy towards the end the experience machines a really good introduction that's another good one Andy Clark talks about active book it's not an active inference book it gives you a much wider view on the uh the cognitive neuroscience and how how the active inference stuff is brought in uh but I really enjoyed that uh and then obviously uh when I'd had this idea and I'd written most of the paper I think I was at maybe you know six and a half 7 thousand words I took the leap and I emailed uh Mr Friston uh and I and I I just emailed Professor Friston with a massive a massive chunk of Humble Pie and said um Professor lovely to meet you I feel like I know you because I've watched you a million times uh online um this is what I've written what do you think and then uh to my surprise um he sent me back uh another 800 words or so um with a very heavy endorsement uh and I couldn't I was a little bit in shock it was one of those moments I was

looking at my computer and I was is that
is that Emil from oh my goodness and
then I opened the manuscript and I was
like wow okay this is amazing um and
then obviously a little bit more
dialogue uh and we chose to uh submit
but needless to say um and I mentioned
this earlier in the podcast because I
don't have a background in maths or
Neuroscience um I'm fly sure that those
holes were evident in that first draft
um and Mr friston did an absolutely
amazing job of of plugging them so I
guess that's the story uh behind the
research Daniel did that did that make
sense

yes you good with that yeah thank
you okay uh have you got any questions
so far or do you want me to just work
through the paper quickly yeah let's
look through the
paper okay cool right I'll just go to
share screen then

um

okay

okay is that Yep looks good there we
okay perfect um so I don't want to uh
read the paper only because I think
everybody will die of boredom but I will
skip through uh the sections and explain
to you my

thinking um so I I go very gently uh
into setting the scene uh talk a little
bit about uh how how we police officers
are often flung into what you would call
Vital High State situations and this is
elaborated on a little bit further but
it's that temporal nature of the

decision-making which I think causes very specific grooves in the police officer's brain and I'm using the word Groove because I think it's a really good analogy that kind of like explains the brain it's not that we can't step out of the groove but they help us make decisions and I suppose that's in line with her istics it's actually in line with like hemispheric Theory like all sorts of like um and I talk about this later like system one system two thinking this is all quite um well established good psychological research it all like kind of melds together it gives you a good little background uh so I talk about a little bit of the history um with regards to uh the hunch and what the hunch is um it's been described using a number of different terms uh so we've got suspicion in here uh This was done on Research that was um conducted when it comes to things like stop and search and stop and search is such a uh a contentious area of policing so uh um high impact when it comes to things like Community tension Community confidence trust safety legitimacy a lot of it is linked with uh the use or the abuse of stop and search so there's been um quite a lot of study that's looked at the decision to stop and the decision to search how those decisions are created and often through um ethnographies where academics have gone out with police officers and watched um uh officers conduct their um their stops and their

searches and how they've uh made they've then asked after that's been done how did you make that decision how have you come to that decision to to stop and search that person what was going on inside your brain and I've then introduced the domestic violence incident which I've already discussed so I'm not going to go through that um but I've just talked about this bit here where it says the scenario creates a fundamental question how does an officer's body generate these hunches these internal alarm bells and what role do they play in navigating the experience of being police officer um now they're not restricted to police officers and I talk about this here especially when it comes to things like um uh medicine uh and Public Service uh based uh professions uh Armed Forces firefighting Etc uh there's uh an excellent article and if I if I could remember it I would share it um but it was about a group of firefighters that went in they're in the midst of like you know uh fighting this crazy big fire and all of a sudden the supervisor just says right get out get out get out and he he grabs all his team and he pulls them out and literally within seconds the floor collapses now obviously there's going to be no uh uh super particular s Salient detail that's made him do that it's going to be a gathering of uh details which are going on often unconsciously in the midst of the fire even him I imagine can't

process

um in in a way that's rational and

deterministic he can't say this led to

this and then this led to this and then

this led to this it just this

overwhelming feeling came to him as a

firefighter and he said no come out come

out quick and as they came out the floor

collapsed

now that's an example of the hunch but

the hunch within a firefighting

circumstance it's the same thing it's

like a kind of overpowering feeling of

intuition uh so I talk a little bit

about this here uh and we talk about it

being a a little bit wider than um

stopping a vehicle um there is a

discussion here of uh um almost like the

spectrum of what a hunch can be like how

does intuition present itself there is

some research coming very shortly from

somebody called Emily Quinn at Cambridge

um she's just done a PhD in something

called The Copper nose which again is

the police hunch and I didn't know about

this um until after i' written it um but

I know that she's done a data collection

I know that she's much more hot on the

terminology and the neuroscience and the

cognitive science behind this and I know

that her research is about to land so uh

just for the listeners it's definitely

coming that's going to be a treat uh

when it lands uh but a hunch or a copper

knows when you're trying to find

vehicles that may be up to no good for

instance uh I'd been in cars with people

who are very experienced police officers

and they say something like I want to stop that car I don't like the look of it now I'm trying to learn and my immediate question would be like well what do you mean you don't like the look of it what is it about that car that's promoting this feeling of unease and even the most experienced police officers they can't answer that question uh they could they'll say something like oh well maybe it's the manner of driving you know but if it was the manner of driving and I might have picked up on it there's something else there something that's that's uh triggering and I imagine their heuristics and their heuristics that have built up over time again it'll be Bayesian uh there will be tiny signs tiny um inputs sensory inputs about a particular vehicle and the way that it's behaving that over time have led to the building up to of this this intuition

now that is a very very small slice of what is intuition a very very big slice of what is intuition might come in the detective world so if you're working on a murder case it could be that you have a very particular feeling about one of the suspects and that feeling may build up over time now again very very difficult to explain why do you why do you have a feeling that that suspect did or did not do it why why is it coming and again those detectives and I know I've worked with them they can't explain it there there's something that's that's building up for them and they can feel

the hunch the hunch isn't something that's visceral it doesn't come into the pyo erection kind of like you know like response to threat or response to stimulus that's severe it's very very slow creeping and it leads to like this it's almost like intrusive thoughts you know like that they're very very slow they come in without they're unbidden there's no like kind of I'm not very very I'm not happy with this you know this is not good you know it's Uninvited uh so the hunch can manifest itself in lots of different ways from the kind of I want to stop that car I don't like the look of it to something that generates over weeks or months and again is very difficult to discern like where does that sens come from we don't know we don't know and that's one of the things that I I think that should be looked at and I discussed at the end when we talk about where the research could go uh so again I take them through um the incident which I've just discussed uh and we talk a little bit about the background of intuition research uh and then we talk about uh rational decision- making intuition there's a meta study there from Wang Wang atal in 2017 he's talking about constructs um I think they are separate they run upon the Dual cognitive system system one system two again emotional thinking what you would call emotional thinking and rational thinking if you were being a Philistine like me

um so yes um and then I I've very very tentatively and I have specifically use that word I'll just highlight it there tentatively with this prior discussion in mind uh the author uh posits a definition of a hunch with the uh for the following discussion which is an often unbidden Salient intuition experienced by a police officer that subsequently influences their actions now when I say influences their actions I don't mean that it it specifically sends them in a particular direction that direction can be anywhere but it's the fact that it causes some sort of change in direction or causes you to consider a change in direction that then uh uh so in other words it's instrumental it's not something that just pops up and it can be dismissed it does something now there was a some uh push back a little bit um on the definition so words were added as we uh went in and through from the reviewers who were amazing uh on the article they were so cool gave lots of different um uh additions or possible additions and talked about this and we ended up with uh this this inclusion of the word salens to kind of gener to describe a hunch you know like what well it's the salience of the hunch it's not like a normal hypothesis our brain is constantly generating hypothesis all the time but there's something about this particular hypothesis when it pops up Wham it hits you okay well what does

that hit feel like where where do it
come from and why is it different to
your normal train of thought what is it
that's that's made it pop
out and that then leads um uh to the
maths now uh some of the narrative
probably halfish of this is mine some of
it calls uh and and Professor friston
went into a lot more detail than me um
the uh equation that I put forward for
surprise was tripled in length um so he
went into a lot more um this was the
basing brain bit we introduced the
basing brain um the if you like the
older research um and then we bring in
the newer research active inference and
free energy
principle uh and then uh Professor
friston uh added the detail in the uh
equations here and also the figures um
which have come in which are the the
background if you like for these
equations how do these equations work
and what do they
say uh but uh uh what's the gist so if I
was to describe
this uh surprisal bounds or the level of
surprise that somebody operates
within a police officer is going to be
very very
sensitive uh to boundaries of surprise
this is the hypothesis essentially
because and I talk about this in the
next section which is about uh cultural
Niche construction because they have
very little control over their
environments and they have very little
knowledge of the people within them

um I I think that their brain will
generate
hunches in a much more sensitive fashion
within those constraints okay so
essentially um uh through the Gathering
of experience uh very uh temporally
informed if you like um work
environment if I go to a domestic and I
walk into a domestic incident and the
male and female are shouting and
throwing things at each other I have I
have no discretion I have to take action
immediately because uh there's there's
harm uh that that can take place at any
point to either of the parties okay so I
have to uh take what is a very active
role in that environment immediately and
therefore I'm constantly training my
brain I think personally as a police
officer this is what I think goes on to
take action and to take action
decisively in order to reach desired
outcomes now in most people's daily
business they have lots of time to make
these decisions and as a police officer
you just don't and that that generates a
a uh grooves I think it generates
grooves in the police officer's brain
and it leads to something which has come
out in the ethnographic literature
around policing called pragmatism now
it's described as a tendency to do over
a tendency to think so a police officer
will often tend I think towards jumping
into some sort a very active role in the
environment sometimes too quickly way
too quickly you know but I think that
that too quickly and I'm going to use

air quotes there is bias I think that
you your brain is trained to
intervene quickly in many many cases and
it sets up grooves and those grooves are
very easy to fall into and that presents
risk you know so um I'll give you an
example if I'm going to be controversial
here but this is where I think the
research will be really cool
if for instance um there is a drug
problem uh within a particular
neighborhood and that neighborhood is
high in ethnic
minority and it could be ethnic minority
from anywhere okay so um it could be uh
uh so in the UK Albanian for instance
okay or it could be black in London or
it could be afro Caribbean there's lots
of different ethnicities but um if there
is a community which is heavy in a
particular ethnicity and as a police
officer you are constantly interacting
in what will be um highly temporal Niche
situations normally violence related uh
with that Community that's going to
influence the way that that officer
thinks and reacts in other um maybe less
temporally influenced um situations so
uh I'll give you a physical example if I
am going to um uh Gang Related violence
a lot okay and I'm dealing with victims
that have been injured as a result of
Gang Related violence if I do a stop and
search in that same
area and I have been to all these
previous incidents there's going to be a
building up of influence and experience
in me which is then going to change the

way that I behave when I stop that car so and you will see this in the research this comes out in a lot of the um uh the way that police officers are perceived when they're dealing with ethnic minorities the way that police officers act they're not the same okay um when it comes to uh majority population interactions okay but I do think and and again I'm pushing this Theory way out from Beyond my paper here I do think that that um there will we can probably find some evidence of why that may be the case uh and and maybe prove empirically that the more incidence that you go to that could be highly temporally influenced that may involve particular ethnic minorities could then Change the way that you as a police officer operate on a much more wider scale with members of that ethnicity now I don't think a lot of that's conscious and I know because I've worked with police officers I I mean I met one racist police officer in my entire service of 18 years so um I know people say are the police officer police police are racist you know I don't think it's as simple as that I think it's way more complicated I think that we have to really understand why police officers make the decisions they do over time for us to properly unpick this problem I don't think it's something that we can turn the page on uh and it's it's going to be years of good research some of which is alluded to in Andy Clark's book

um so anyway I'm getting a little bit wide here but all of this uh influences that conversation that I've just had I'm not gonna go into the detail but that's that's that's the bit that that is there um and we have this wonderful addition from him uh which gives you the background to the theory uh from Professor

friston uh now this this is the section that if I had the choice I would have put a lot more um words into uh and this is the hunches and nce construction stuff and you'll probably be able to tell that from how I've been talking about it today um but low surprise longitudinal relationships probably make up the vast amount of our uh interpersonal relationships that we all experience on a day-to-day basis all of us um in general in the world unless you're in somewhere that's very very volatile mostly they're going to be low surprise longitudinal relationships so you're going to have more opportunity to mold the environment over time okay uh so that's the and that's the active bit of the inference okay so you've got this like kind of ability to interact with the environment the environment then interacts with you and you end up with this like kind of constantly like forward and back in between you and the environment the problem is of course uh again with police officers the ability to longitudinally uh change or influence an environment is just not there you have I have an hour at most at the

incident and then I'm getting sent to another and that may then cause I think this pragmatic lean into to do rather than think so that's the kind of like the the the what I'm getting to in this last this last part here before moving into the discussion so um there's a couple of questions that I've come up here I've mentioned the pragmatism here from Loftus this is from the cultural literature so um what cognitive signals would uh were generated by IAL attendance that contributed to the hunch generation when the basian brain processes these signals what is produced empirically during the process of active inference what was the officer's qualia so how does the hunt present into Consciousness massively complicated and I've even put it in the paper we don't want to get into the the Consciousness debate because it's absolutely fascinating but it's just incredibly complicated so there are some things here that make that person aware and this refers to that salience which is in the definition how were those qualer interpretated so when the hairs on my arm stood up how did I see the hairs on my arm standing up so obviously you're going to receive the signal but how do you interpret and then turn that signal into action these are questions that haven't really been answered yet within this particular area and then um this last one which I think I mean

it looks like an afterthought because
it's one sentence but it's not one
sentence it's actually really important
how did the officer justify this
hunching latest Court proceedings and I
have written hundreds probably thousands
of police statements over the years and
I know that reading back those
statements they will read very
rationally they would read super
rationally so my ability to rationalize
my decision- making post decision making
is way better than my ability to
interpret a hunch rationally and there's
a big disconnect there so um uh you end
up with a lot of obviously hindsight
virus if you stopped a car and you found
some heroin well you stop the car
because you think that there was drugs
on the car but as mentioned earlier in
the um article sometimes that hunch is
non-specific it's raised through
experience that's that's very very diff
different really different and I don't
think the officers have got the skills
to interpret the hunches well enough to
make well reasoned decisions I think
that they rely upon um intuition much
more than they say they do and I know
well I know because I've been one and
I've been around them I know how these
things work so I just don't think that
it's been researched
well um and then as we get towards the
end um uh we move into the again the the
theory uh support from that figure and
we posit a series of areas for future
research so we've got here long Udin

study on hunch development so this is tracking the development of hunches in the in police offices as their career develops I think this is super interesting because when I talked about going up to that door in the domestic and Jamie not feeling anything and me having the most Salient feeling I've ever had there's a there's something going on there you know like so when does that start to happen like when do when do Police intuitions become so noticeable and Powerful that people can see them and be able to act upon them will be able to notice them and suppress them because even that is an interesting area so if you are getting a hunch and you think that hunch may be informed through past discriminations then surely you need to be able to suppress it you know you shouldn't be acting upon it um so if you think for instance that um uh there's an element of discrimination in stopping a vehicle you need to be able to notice that and stop that before it happens you know and it could be unbidden and again this is in the definition so you haven't invited the hunch it just comes but you may know that there's not enough there for you to make that stop there's not enough evidence so something's going on and you need to be able to see that and stop that happening uh comparative analysis with other professions so this is looking at the generation of hunches for instance in the medical environment

neurobiological correlat so this was to look at um neurobiological processes associated with hunch formation uh the use of brain and body Imaging techniques this is probably way above my pay grade but it would be super useful and there is some precedent for it in very very primitive studies in the 70s and 80s that look at things like um heart rate and blood pressure over the Tour of Duty of a police officer you know um but it'd be brilliant to see that Advance much more advanced wearables even looking at good scanning equipment how are these countries generated it might even be one for the future but we thought we'd discuss it um the impact of training on hunch accuracy this is something that came from Professor kpes he was talking about training scars um have you heard of training scars Daniel no so training scars are where you constantly train in a particular way your grooves become very very pronounced in the way that you approach similar incidents because you train so much and then when an incident presents itself your brain kicks you into those grooves even though the grooves may be inappropriate so they can lead to um uh for instance taking a shot because that's a shot you would have taken in training despite the fact that that shot was non-justified or another one where there's not enough threat in the training scenarios so because of that you act in a much more uh a slow and considered way when you need to act much more quickly

both of those will be considered
training scars and again um uh so
overtraining may suppress good hunch
Intuition or development
that's the an idea I'm not saying it's
there it's just an
idea uh so diversity and we've discussed
this quite a bit today um insight into
potentially gendered hunches I mean I'm
super interested in that I might look
this plays out when you look at the
stats and things like stopping Vehicles
it is different it's different from a
agenda basis there's something going on
no one's looking at it I think this this
needs looking at um oper
operationalization of hes and policing
so

um when and I'm currently planning a
study that looks at this now so using
both qualitative and quantitative
methods investigating the factors that
contribute to the activation and follow
through of hunches but to do that you
have to be aware that a hunch is
happening and that's a that's the one of
the things that we're we're trying to
get our heads around like how do we how
do we make officers aware enough that
intuition is kicking in and then when
that intuition is kicking in how do we
help them describe it and also help
describe their decision making off the
back of it because I think again if we
go back to that pragmatism side I think
the hunch kicks in the police officer
makes the decision but there's very
little rational consideration of the

connection between the hunch and the action so that's something that I think we can look at um compar comparative analysis of policy models and I'll give you a really quick and dirty example of this so in the US obviously um it's a coercive policing model in the UK it's a consensual policing model both of those will change the way that hunches are generated they will um so have you heard of the touching the tail light on on police vehicle so in the US when um a cop touches uh stops a car as the police officer walks towards the car to speak to the driver they touch the tail light now there's lots of reasons for this the first is that you leave your fingerprint and DNA on the tail line okay but if you think about what that's doing in the officer's brain bra the immediate consideration when he gets out of that car is I could die and I need to leave evidence on this vehicle in front of me now I have stopped thousands of vehicles in my police career and I have never once thought that and because I've never once thought it I can hypothesize that the generation of hunches will be way different in terms of sensitivity to an American police officer's generation of hes not least because the threat is so mitigated in the UK the chances of me coming up against a firearm is almost nil I have never seen a gun I've never fired one until about six months ago so like 18 years of service and I've never carried and I've never seen a gun in a real life policing environment that's

going to change the way that I police
and I do think that there'll be a
different operationalization of hunches
as a result and that's something that
would be a fascinating
study and then um this one here this
impact of technological support how does
the use of Technology um change hunch
development so for instance and this is
going to sound really weird but let's
just say a police officer is trained to
record when a hunch is received and it
could be as simple as a tap on the wrist
for a wearable or something
similar and uh that could then help us
correlate the types of incidents that
police officers are at when they receive
hunches they don't have to explain them
it's just like what where are you most
likely to get them where does your
intuition manifest you know and then we
could look backwards at how it
manifested and what it did and then the
last one which is the public perception
and Trust um possible to investigate how
public perception and Trust in law
enforcement are influenced by officers
Reliance on hunches so I'll give you a
quick example of this if you rely on
hunches to do stop
searches and that means that you do more
stop
searches that may affect Community Trust
within that uh that particular community
where you're conducting the stop
searches that then becomes reciprocal so
you rely on intuition to do stop
searches rather than rational thought

that leads to a rise in stop searches
leads to a rise in arrests and also
probably a rise in much more like um
friction based interactions with members
of the public long term that's going to
lead to a breakdown in in relationships
and possibly um issues with legitimacy
so how the public perceive um uh the
legitimacy of police
actions so that's just a a an example of
where we could go on the big scale this
last one's a very macro like kind of
level study and some of these up here
are much more personally
based and Daniel I think that's about
it thank you thanks for overviewing
it I have some questions and there's
some great questions in the live chat so
okay that sounds
good wow many places to jump in there's
the fundamental tension with
the possibly arbitrary or hard to
explain yet intuitive and
honed nature of the
hunch and about how the hunch has such a
rich before during and after there's
kind of there must be something like
hearing the code called on the radio
yeah and that setting a stage like I'm
going to a comedy I'm going to a horror
movie or it's this code and that's all
that is being passed through the blanket
is just what code it is and that's
equivalent to like saying what is your
prior what do you prepare yourself for
given that it's just this number or that
number and what are the costs and the
sensitivities to like false positives

and false negatives
so there's a lot of stages to
investigate like looking forward to the
Hun then looking back on it where
certain kinds of rhetoric or certain
kinds of thinking are privileged or
supported by legal
proceedings and saying something like
well I just didn't think it was that way
in the moment could be the most honest
statement yet it might be seen as like
almost
malpractice okay awesome questions from
live chat yes exactly
right okay all right here we go I'll
just read them um basically in the order
they in all right so Susan asks does
hunch lead to epistemic foraging or
pragmatic foraging so let's just unpack
that in the expected free
energy decision making
functional different policies are
evaluated in terms of their epistemic
value like expected Information Gain and
pragmatic value which is aligning
observations with your preferences so
how does that raising of the
hunch then get pushed one way or another
into seeking more information
epistemically or directly moving to take
pragmatic
action so I think that was a a very nice
way of summing up a lot of the
discussion that where I AI about the
lean towards
pragmatism the desire for control for a
police officer when they first get there
is like it's almost Primal and I know

that's a big word to use um but they will I think tend to lean incredibly strongly into the pragmatic and then move into the epistemic so and if you think about this this mirrors the court process like if I if I turn up to an incident and there's threat within the incident the first thing that I want to do is and I know the word is awful because of the connotations it brings but uh neutralize the threat I want the threat gone because while the threat is there uh my intuition is going nuts I mean it overrides everything if I'm if I'm so I'll give you a very quick a practical example if I'm in an incident where I believe that the male party in domestic violence is going to assault me or assault the victim while I'm there right the my brain does not want to allow any epistemic foraging none there's a safety override and the safety override is essentially make sure that this male is not going to assault you or assault the other person before we start asking questions and that safety override is often the thing that people will look at and say and this is what I meant when I talked about the difference between jurisdictions so I would say that the safety override in the US is much more powerful than it is in the UK because of the threat that's there because of the availability of firearms so you're probably going to find where there's more Firearms the link towards um uh pragmatic verigin is way higher way way higher and then once that's done

neutralized whatever word you want to use then that's when the the longer the longer form questions and the much more like calm information gathering takes place so really good question there's definitely a lean uh towards pragmatic over epistemic but epistemic will happen and it just may come after the pragmatic so there could be a like kind of hierarchy uh a hierarchy if you like um performances because of the situation that presents does that make sense yeah like to give a kind of a non policing example it's like you dropped an object on the floor if it was very surely just soft carpet you might feel around blindly and epistemic foraging would have like a low risk but if you strongly believed that there was like other sharp objects on the floor you would be way more skeptical with just kind of reaching into dangerous locations that is Daniel can you answer my questions because that was way better than I just did okay next one we'll each give a shot we'll each give a shot um Susan asks how are police trained to avoid group think I.E not just follow the ones with improper force over Mor motivations So you you're talking about individual cognition in that moment but yet in that diad encounter there was like a collective intelligence component and a difference in hunches so how does the group training and operating come into play uh yeah this is really interesting

because there's nothing about the way
that your brain behaves under pressure
in any of the police
training
nothing so and I suppose and this is a
massive criticism and I and I used to
talk about this a lot because I was
involved in the design of What's called
the pqr in the UK which is the education
qualification
framework um and it was a very very
controversial project for me to be
involved in but it was about raising the
initial level of Education to degree
level bachelor's level for police
officers and one of the major drives is
obviously to blunt to some extent the
lean towards pragmatism and offer some
prayers for the making of police
decisions that are better than those
based just in practice but this is a
move away from what you would call like
a kind of Master Apprentice model which
is the uh and its experience-based
epistemic model now of police learning
into something that's much more a priori
which is um your more traditional
educative professionalized model so are
they taught to ignore group thing I'd
probably say well there was no
discussion of group thing during my
training but I have to think back I mean
my training was 23 24 years ago and I'm
sure that things have
progressed will it be good inform formed
solidly evident stuff probably not and
um I think in some cases I think there
may be some really good academics who

will be out there working with particular forces and they will be giving this sort of training but I think as a more General approach I don't think it's there and I think it's a massive Gap so really good question all right yeah very interesting it's um it's a team sport team art SL science and so it has to be at least considered how do do officers share their hunches do they just raise their hand and say I have a hunch but that might be throwing a wrench into things in the moment um so H how do the right uh communication surface and then you have the the radio and you have the local communication you have these different Avenues of message passing but not all messages lead to resolution of the situation

So Daniel that's a really little comment um we just submitted a paper which is on uh the use of decision making models in detective world and we were talking about this like I was only on a murder inquiry for a year of my service so um but I saw the way that the murder inquiries worked and it is way more rational and they invite hunches so like your senior investigating officers will say look have any of you got any feelings about this please try and explain them you know how or the or the other one where you just run an interview with one of the suspects and one of the first questions that would come out isn't what did the suspect say

it's what do you think how did it make
you feel you know so there's a a a real
space within the detective where world
for the discussion and sharing of
hunches now if I pick that same um uh
instance up and you drop it into uniform
because it's so temporarily enforced
that you won't get Chance a lot of the
time to share your hunches like so if I
have a bad feeling about somebody at an
incident and this is that lean towards
pragmatism again I will want to make
sure that I'm safe before I go anywhere
I won't share with the officer next to
me I have a bad feeling about that
person I want to take pragmatic action
to go and try and deal with that bad
feeling so again this leans back into
that other question that we've just
discussed I would say that so the
sharing of hunches and the passing of
messages is probably much more available
within the long the
long decision making processes within
detective work and very poor within the
incredibly short temporally influenced
decision- making of uniform does that
make
sense yeah it it also makes me think
about active inference and statistics
are providing like a science of
uncertainty and decision-making amidst
uncertainty and yet in many different
domains me ranging from policing to law
science there's kind of bespoke ways of
dealing with
uncertainties and different settings
even within the given field like

different cases are going to have
different treatments of of uncertainties
and so there's now this dance or Fusion
to bring together those different
treatments of
uncertainty um I'm really curious how
you see active inference and extended
cognition in light
of even though I'm not very familiar
with the details the growth of policing
Technologies and surveillance
Technologies and the different paths
that a given officer or Department might
have in front of them in terms of how
they incorporate
technology into their whole
apparatus so there's definitely going to
be a really really big shift here um so
when I left so this was two and a half
years ago there were discussions three
years ago about things like the
integration of active headsup
displays okay so when I'm at an
incident if I've got facial recognition
and it's running in the background and
it's passing me information that I don't
have and didn't have um how will that
affect my decision making and that is
and I'm going to be really honest really
scary uh
because really scary and really
beneficial so I'll give you a quick
example I turn up at an incident again
I'm I'm going to keep using the domestic
incident because it's simple and we and
we as police officers go to them all the
time okay so it's a and I keep calling
myself a police officer I'm not a police

officer anymore I'm full-time academic
but you know long time so anyway um if I
turn up at an incident and I have set of
glasses in my body cam on my body Cam's
running at all times so I know that
there's some element of surveillance of
my behavior and I know that whenever you
look back at this incident my decision
making is going to be um scrutinized
okay so when it go when it goes into the
court system my behavior will definitely
be on show and I know that as a result
obviously I'm I have to behave within
the rules and the bounds and all that
sort of

thing now there's a propensity and I'm
sure you know this for police officers
to do what's called informal working
practices where they essentially step
outside the strict rules and laws and
they try and take pragmatic decisions so
um uh an example might be if I stop
somebody speeding I might want to give
them a warning I might not want to give
them a ticket but speeding is
technically a strict liability offense
so I should have to give them a ticket
and instead I operate and I use
discretion I I decide not to do it I
give words of advice and that person
drives off

okay with things like your heads up
display if I'm walking towards that car
and I'm wearing glasses that has given
me full history full driving history
including infraction speeding tickets
Etc just by looking at the number plate
and then when I get up to that driver's

side window and the window comes down
and my facial recognition tells me that
this person has got previous for x y and
Zed ass Soul Etc is that going to change
my decision- making 100% because what
what has the police officer got to deal
with now they they have what's in front
of them and then they do that epistemic
foraging the extended cognition that
would come from Lifetime information
during decision making is crazy um and
I'll tell you right now we'll be doing
it before we understand whether it is a
good thing to do or not because that's
what that's what police do they will
jump in probably in fact I'd be
surprised if people weren't testing this
stuff now um uh

and and the other thing I think I mean
I'm just going I went straight to
wearable stuff there but there's other
stuff that might help so if I'm thinking
from an interceptive perspective there's
a really good study

um called emotional survival for law
enforcement officers and it was a psych
it was a police psychologist that wrote
it and he put heart rate monitors on
police

officers and he showed that when you go
into What's called the cold one incident
in the UK which is an eer mer incident
the blue lights are on you know like you
were talking about the Cod the passing
of the information and the the bodily
reactions to it if I'm going to an
emergency incident when I first start
you're incredibly aware intercept ofely

of what's going on so you can feel your heart you you your um breath quickens your uh periphery Narrows all of the normal stress um uh uh reactions kick in okay because you know that when you land there there's probably going to be conflict of some sort and you're going to have to deal with it now what the study showed was that very very experienced police officers so we're talking like 15 years 20 years plus when they were going to emergency incidents they would say that those bodily reactions were not happening but the um monitors would say that they were and this is super interesting because it may mean you you could say that the police officer has got more control over their interception or you could say the exact opposite so they have dulled their interception over time uh through exposure continued exposure and we don't know about the effects on hunches or the effects on General decision making in those different lengths of service if you like now if we go back to the extended stuff imagine having a wearable that tells you this lifetime that would be super interesting like it I mean again I'm like you know maybe 5 years in the future here but we're not a Million Miles Away there are well-being variables that are ongoing now I don't think they're being used for this purpose could they change decision making definitely will they and in good and bad ways hey Daniel this just needs

a lot more research this bit this bit
that you're discussing loads more
research it just needs some proper
proper good looking at uh because it
could be immensely beneficial I can just
see a million ways where it could also
be quite damaging
wow yeah the the awareness of the hunch
and then there's one setting where the
hunch is not aroused which may be like a
burnout situation yeah exactly and then
yet also on the other side of let's just
say like a stage performer they still
might get the physiological arousal but
they actually expect it and so they
don't attend to it because this is the
whole thing with
we're not paying attention to what we
expect yes and so they have a deeper
experience layer that contextualizes the
hunch and yet that might
obscure that hunch from being
communicated like to a younger colleague
yeah and yet communicating where and how
the experienced Professor or lawyer or
officer feels in a situation is that
ineffable experience transfer and then I
I totally see these different paths with
the wearables One path where the um
boots on the ground are just reduced to
an actuator in a decision-making system
that they're not even a part of because
it tells them here's what you want to do
to this car don't epistemically forage
just fulfill what the order is exactly
and then there's the epistemic
engagement mode and we can imagine like
the family doctor asking somebody

questions rather than like looking at their digital health record and there's kind of um there's a charming and an open-endedness to that and yet in high reliability situations people are going to be put into which lane are we going to take this doctor's office are we going to use digital health records or are we going to have slower potentially noisier encounter just because people self-report that they like that more than the other way yes yeah and just as you got me thinking there there's some really good studies that look at things like um uh cancer diagnosis and so you know somebody comes in with a bad ankle and a doctor says yeah you need to go for this test and this test and this test and they're unable to rationalize that it's the same thing but there's a much more so if you look at a doctor's purpose there they're still going to be pragmatically B they're going to be trying to solve whatever problem is facing them um but their relevance realization has got so skilled over the years and this some of the stuff that you were just talking about there where there's lots of signals coming in that are just very very efficiently dealt with subconsciously and it may be about making that person aware of what that subconscious is doing in order to properly understand it and this is what I was talking about with the hunch Generation stuff because police officers don't often they don't interrogate the

journey that led them to make the
decision they make the decision and then
they rationalize it
backwards I think there's so much we
could do here like there's so much
potential because and again this is for
improved decision making this isn't like
um so we're looking at improved outcomes
here we want to make policing better how
do we make it better well I would say
that increasing self-awareness about
decision- making is absolutely
fundamental you can't go anywhere this
is just like perfect you know and the
whole like the whole framework of active
inference and
um and the basing brain it just lends
itself to this so well way better than
the other theories that that I'd
explored and some of which are in the
article like you know like the schema
the scripts Etc um but I love ramstead
work on nich construction in it that's
something I think that I would like to
look at a little bit more I think as as
time goes on sorry my brain just went
there towards the end I don't know why
yeah I mean is there anything else than
your question why have we got some more
yeah yeah okay let's keep going with the
questions um Jeff
asks what overlap exists with military
psychology zilon identifies active
inference without directly referencing
it in his recent book beyond the pale
published by air University press I
don't know the book but what do you
think of the general question uh I don't

but I have um so I work in a a security
and defense uh Academy so I'm surrounded
by uh military Scholars uh that's a
really good question and I will go and
ask it for

Jeff cool okay Dave asks how do hunches
align with hallucinations in deep
learning

[Music]

oh
oh that's a really good
question um and I have to be honest I
mean I've read some stuff on
hallucinations and I've listened to
podcasts from machine learning Street
talkers as well as some good good
discussions around

it I mean they're anomalies aren't they
but and I suppose you're going to have a
set of rules within the llm that still
allows them to be displayed played in
the same way that you're can have a set
of rules within the body that allows the
hes to come to the

four I don't know is the answer to that
um and I know that we don't properly
understand hallucinations yet but I
would I mean obviously we've got no
consciousness in in in the the Deep
learning machines point of contention
but most people would say there's no
consciousness there whereas the
hunch the hunch is ability to rise
through the Consciousness so the qualer
of the hunch is the bit that makes it
Salient and I think that that salience
would be lacking in in any element of
llm deep learning based work if that

makes sense does that make sense
Daniel yeah I I think uh another
complimentary way that I thought of that
question is the
hallucinations let's just say in image
output you say represent this and it
makes a full resolution image that is
non-f factually based and so people
sometimes refer to it as
hallucination so whereas the outputs of
that are in terms of
observations that are just factually
ungrounded the hunch it's not like it's
a it's not a
visual phenomena it's not like there's a
visual a new visual overlay or a sensor
prediction it's actually something
slower that is kind of like an upwelling
of something that is Salient because as
you pointed to it's slower it's
interceptive it's affective it's not
modifying like what we see smell hear in
the way where when people talk about
hallucinations as outputs of machine
learning models usually might do that
though Daniel that sounds really weird
but I think it may to some extent do
that so um and I'm just thinking about
how that might be the case so if I'm at
an incident I I'll give I'll give you
another very quick example so uh we went
to um conduct a mandatory section in
which is a horrible incident and I hate
them but it's essentially taking
somebody into full-time mental health
care he was a serious um schizophrenic
with he was a danger to others um and we
went to uh basically take him into

custody so that he could go to the hospital um because he wasn't he wasn't refusing to take his medication or short for his appointment and it was caring the community Etc and he had attack somebody and I was in that room for probably one minute and I was telling the doctors to get out because I knew what was coming and if somebody had and this is what you're saying well it changes the way that you see well it does because it changes your relevance realization is that as the intuition kicks in and becomes stronger then your focus changes visually so you're going to be filtering out much more of the the side if you like and focusing in on what be what be considered the threat whatever the threat may be um and in this case I mean he was holding a knife down the side of a sofa and he was about to attack us and for some reason something about that incident again highly Salient feeling it then changed the way that I saw it changed the way that I interacted communicated changed the way that I dealt with my externals um uh and and I suppose you go call them variables so I wanted the other people out of the way because that would then give me greater uh Le that would give me less to think about when the conflict actually happened so I would probably say there's a little there's a parallel there I think that it can affect the senses and I imagine that no I quite like it it's an

interesting thing to think about
that's I saw in the paper you cited the
the quantum reference frames and Chris
fields's work so that just makes me
think walking down a street one
reference frame is just like oh there's
people Milling about and then another
reference frame whether this is paranoid
or not is like their information to
Circle

me or somebody is shaking a little bit
in their hand and it's like oh maybe
they have essential tremor maybe they're
nervous maybe they're about to do this
and so that doesn't

change the the photons hitting the
retina that's right it changes our

Framing and that is what is
actionable and a more cognitively
informed law and legal

system could have a more
agreed right grasp of this yes you're
exactly right yeah yes very good I told
you you're way better at this Daniel
like you wrap up my words and it just
comes out in a in a in a much simpler
way very good let's we're we're cruising
together um well where are you going to
go in your continuing directions
here like what are you going to do with
this now that you have the

paper okay so um there's a couple of
things that I want to do I want
to uh

test um whether current decision- making
models uh in policing that are used
widely in many different
countries um are useful for the

processing and uh understanding of police hunches because I think that they are accountability models and not decision-making models so that's something that I want to look at but I suppose the more uh relevant study will be researching the uh qualia of the hunch so um within different settings especially within uniform I'm very interested in it because of its its contact with legitimacy as a whole like how police perceived I want to know how uniformed officers receive understand process and then act upon punches uh and uh the study that I have in mind is this is going to sound really weird um I want to do like a kind of personalized dictaphone like a narrative account so essentially when a police officer in any corner of the world makes a decision on the basis of a hunch that is then narrated into this kind of like um Repository and we use that repository as a a qualitative um evidence-base to research how uh the hunch is process and then acted upon that's what I'm thinking so um I'm currently in the um process of getting funding and uh I'm hoping that I can secure it and if I can I'd like to run it as a this is going to sound really strange but a voluntary worldwide study so I don't care where you're a police officer um I I want you to take part uh and we look at uh how you are receiving on a um a daily basis your minor hunches from I want to stop that car through to those like kind of heart

hitting your stomach pyo erection High
threat hunches that totally change the
way that you see the incident in front
of you so that's the that's the next
step I think that makes me think of
aroid which is a repository of alternate
states of consciousness mostly based
upon
substances pyel yeah it's another but
it's a it's a voluntary self
report and that sounds very cool and
voluntary I'm hoping I'm hoping that's
the plan but um I'm hoping that once I
get the funding that that will be a
green light and obviously in the
meantime I mean it's been so cool
talking to you Daniel and I I know that
the community is quite tight in around
the active um uh INF Institute uh so if
anybody has been listening or if anybody
listens to this in the meantime if
you've got any cool ideas or you want to
ask me any questions look just email me
I I I love to talk about this stuff in
general and the discussion I know that
we mentioned it right at the beginning
but even the discussion where we talked
about the law I've got I've got a theory
that I've spoken to a professor about
I'll just throw this out there just to
finish
off I think that people may call the
police when the
surprisal bounds are breached and they
want the police to be an agent of return
so essentially The Police become an
agent of AI estasis to return to home
estasis for that person now what that

gives us and it's something that we don't have is a general theory of police contact and it allows us to see how different communities may call the police based on their different surprisal bounds so for instance in high chaos community is their sensitivity to surprise is going to be much higher than those that are in um sorry much lower than those that are in um uh very very structured gated communities for instance um and this this could lead us to some really interesting research but it's something that is very much in the background of my brain and something that I'm I'm I'm thinking about a lot reading about a lot and hoping that maybe over time I can start to develop awesome thank you Gareth for um connecting these critical areas and bringing active in and it's just exciting to hear where it's going oh thank you Daniel thank you for your time and thank you to everyone who's actually managed to listen to me to drone along for this long it's it's been a pleasure to have you oh there's great comments in the live chat so thank you talk to you next time okay bye thank you very much bye bye see you for